

1. Solve the system of equations by any convenient method, if a solution exists.

$$\begin{cases} 4x + 5y = 7 \\ x = 6y + 4 \end{cases}$$

Select the correct answer below and, if necessary, fill in answer boxes to complete your choice.

- ☐ A. The solution is $x =$ and $y =$. (Type integers or simplified fractions.)
- ☐ B. There are infinitely many solutions.
- ☐ C. There is no solution.

2. Solve the system of equations by any convenient method, if a solution exists.

$$\begin{cases} 3x + 2y = 0 \\ 9x + 6y = 2 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. The solution is $x =$, $y =$. (Simplify your answers.)
- ☐ B. There are infinitely many solutions.
- ☐ C. There is no solution.

- *3. Solve the system using the method of your choice.

$$\begin{aligned} 0.4x + 0.3y &= 0.9 \\ 0.7x + 0.6y &= 1.6 \end{aligned}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- ☐ A. The system has a single solution. The solution set is $\{$ $\}$. (Type an ordered pair. Simplify your answer.)
- ☐ B. There are infinitely many solutions and the equations are dependent. The solution set is $\{(x,y)|0.4x + 0.3y = 0.9\}$.
- ☐ C. The system is inconsistent. The solution set is the empty set.

- *4. Solve the following system of equations using matrices (row operations). If the system has no solution, say that it is inconsistent.

$$\begin{cases} x + 2y = 8 \\ 3x + 6y = 24 \end{cases}$$

Select the correct choice below and, if necessary, fill in the answer box(es) to complete your choice.

- ☐ A. The solution is $($ $,$ $)$. (Simplify your answers.)
- ☐ B. There are infinitely many solutions. The solution can be written as $\{(x,y)|x =$, y is any real number $\}$. (Simplify your answer. Type an expression using y as the variable.)
- ☐ C. The system is inconsistent.